

Jonathan Pan

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Education

University of California, Irvine

Expected June 2027

B.S. in Computer Science

GPA: 3.81/4.0

- **Coursework:** Data Structures & Algorithms, AI/ML, Data Mining, Database Management, Computer Vision

Experience

Software Engineering Intern

June – Sept 2026

MyCase (8am)

San Diego, CA

- Built **Documents QuickView**, a document previewer in **Rails** and **React**, cutting p95 latency to 1.7 seconds.
- Replaced **pdf.js** range fetching with a size-tiered **S3** download, reducing PDF fetches from hundreds of requests to four.
- Designed **Mixpanel** telemetry for preview performance, capturing split timings and failure causes on every settled attempt.
- Shipped **TypeScript** APIs for a **tiptap** legal editor, letting hosts enable or disable formatting controls.

Software Engineering Intern

Sept 2025 – June 2026

Canon U.S.A. – Future Services

Irvine, CA

- Built a **GCP** document platform using **Gemini on Vertex AI** to classify and extract data for **5 production tenants**.
- Shipped **pessimistic document locking**: lease-based **Firestore** locks synced live via **WebSockets**.
- Engineered the AI classification and extraction pipeline to a **<8% error rate** across **50+ document types**.
- Cut document cold-load latency from **30s to 14s** by deferring page rendering and parallelizing AI classification.
- Integrated the platform with Canon's enterprise document management system, shipping reliable large-file uploads.

Software Engineering Intern

June – Sept 2025

Canon U.S.A. – Service Automation Tools

Irvine, CA

- Developed an AI (**RAG**) support chatbot on **Azure** with over **300K+** service manuals and device specific diagnostics.
- Reduced AI inference costs by **40%** through document filtering and prompt compression.
- Tuned retrieval and prompts through **A/B tests**, validating changes against technician resolution outcomes.

Student Researcher

April 2024 – Present

U.C. Irvine – BIASLab

Irvine, CA

- Engineered a **PyTorch CycleGAN** for privacy-preserving object detection, retaining **87%** of original accuracy (mAP).
- Designed a spectral decoding framework for **audio language models**, reducing input token counts.
- Created and annotated a **2,000+** image dataset for evaluating privacy-preserving computer vision models.

Software Engineering Intern

July – August 2024

Altek Medical

Hsinchu, Taiwan

- Built a **Java** DICOM pipeline for 1080p video, reducing latency to **<200ms** per frame to support real-time data streaming.
- Implemented **TLS** encryption to securely archive patient media and metadata in a **PostgreSQL** PACS server.

Leadership

Projects Chair | Artificial Intelligence at UCI

July 2024 – Present

- Authored an ML tutorial series (linear classifiers → **CNNs**) and mentored students on projects in **Python/PyTorch**.
- Taught members to build **multi-agent workflows** for coding with verification loops and multi-role decomposition.

Projects

Street-Art | Python, Node.js, Tailwind, Firebase, Git

- Built a web app transforming uploaded images into navigable GPS routes (“GPS Art”) for Strava runners and cyclists.
- Implemented **Iterative Closest Point (ICP)** to align sketches with street topology and **NetworkX** for route optimization.

Vroommates | Python, React, Flask, Firestore, Google Cloud, Vercel, Git

- Built a carpool platform matching drivers and riders via route algorithms with **Google Maps**, Flask, and React.
- Persisted ride, rider, and match state in **Firestore**, supporting concurrent users across sessions and devices.

Skills

Languages: Python, TypeScript, JavaScript, Ruby, Java, C++, SQL

AI/ML: Vertex AI (Gemini), PyTorch, LangGraph, RAG/GraphRAG (Neo4j), OpenAI embeddings

Cloud/DBs: GCP (Cloud Run, Firestore, GCS), AWS (S3, Lambda), Azure, PostgreSQL, Redis, Terraform (IaC), Docker

Frameworks/Tools: Rails, React, FastAPI, Node.js, Next.js, Jest, Vitest, Playwright, GitHub Actions CI/CD